Primary Manuscript Values Are Not Enough: Situational Modifiers Shape Value-Based Decisions in LLMs and Humans

Key Highlights

- Research Problem: Values alone do not predict moral choices; situational modifiers can shift decisions in humans and LLMs.
- Proposed Solution: Create VISDA (80 scenarios, 8 modifier axes) and a decision-flip metric to measure context-induced choice changes across value profiles.
- Unique Contribution: Evaluate 3 LLMs, a utility baseline (95 Schwartz profiles), and humans; reveal LLM miscalibration favoring personal-cost modifiers.

Summary

Title

Values Are Not Enough: Situational Modifiers Shape Value-Based Decisions in LLMs and Humans

Author List

Anonymous ACL submission

Research Overview

Introduction

- Personal values are often considered stable principles guiding behavior across situations, with Schwartz's value theory organizing values along dimensions such as openness-to-change versus conservation and self-enhancement versus self-transcendence.
- Schwartz values are frequently used to explain differences in moral and social decision-making, mapping more directly onto action-level trade-offs compared to broader frameworks like Haidt's moral foundations theory.
- Recent work in psychology and LLM research often assumes that a known value profile reliably predicts behavior, but situational context can strongly alter decisions even when values remain fixed.
- Decades of social psychology suggest that behavior depends not only on stable traits but also on immediate contextual pressures, indicating that values alone may not fully predict behavior.

Key Research Problems

- The research paper addresses the problem that personal values alone cannot fully explain moral decisions, as situational pressures can shift actions even when values are fixed.
- Current LLM evaluations of value-driven moral reasoning inadequately capture situational context, leaving it unclear whether LLMs act as stable value-driven agents or exhibit human-like context sensitivity.

Methodology and Solution

- The paper introduces VISDA (Value-Informed Scenario-Driven Actions), a benchmark of 80 binary moral scenarios with systematically designed situational modifiers.
- Each scenario pairs two actions with modifier sentences, and situation-driven change is measured via a decision-flip metric.
- The VISDA dataset consists of five structured components: themes, scenarios, modifiers, value profiles, and profile descriptions.
- Decision-making is modeled through a utility-maximization baseline defined over the Schwartz value space, incorporating contextual influence through additive perturbation vectors.
- Three open-weight instruction-tuned LLMs are evaluated: LLaMA 3.1 8B-Instruct, Gemma 4 31B-Instruct, and Qwen 2.5 32B-Instruct, using greedy decoding for deterministic outputs.

Experiments

- The paper evaluates three open-weight LLMs and a utility-based mathematical baseline over 95 Schwartz profiles and a human pilot.
- Each modified trial is paired with its corresponding baseline trial to compute a within-subject decision flip, yielding 7,600 paired comparisons per model.
- The human study collected value profiles and decisions from 50 participants through a questionnaire, contributing approximately 500 decisions.
- Decision-flip rates are computed for all systems, with detailed analysis presented in the paper.

Key Findings and Conclusion

- Situational modifiers substantially alter decisions across systems, with LLMs and humans showing nontrivial flip rates.
- LLMs detect that modifiers matter but redistribute pressure across modifier axes differently from humans, indicating axis-level miscalibration.
- Humans are more responsive to stakes and affective modifiers than current LLMs, which overweight personal-cost modifiers.
- The paper concludes that personal values are necessary but not sufficient predictors of moral behavior in both humans and LLMs, exposing a directional miscalibration in current open-weight LLMs.

Key Concepts & Notations

Important Concepts

- Schwartz's Value Theory: A psychological framework that organizes human values along higher-order dimensions such as openness-to-change versus conservation and self-enhancement versus self-transcendence. It is used to explain differences in moral and social decision-making.
- Situational Modifiers: Contextual factors that can alter decisions even when personal values remain fixed. These modifiers can include pressures such as authority signals, self-preservation, resource scarcity, and social visibility.
- Value-Informed Scenario-Driven Actions (VISDA): A dataset consisting of 80 binary moral scenarios across five themes, each augmented by eight situational modifier axes, used to study the effect of situational pressures on decision-making.
- Decision Flip: A metric used to measure situation-driven behavioral change, defined as a change in the selected action after introducing a situational modifier while keeping the underlying value profile fixed.
- Utility-Based Mathematical Baseline: A model that predicts decisions based on a utility-maximization approach over Schwartz value profiles, incorporating situational modifiers as perturbation vectors.
- Portrait Values Questionnaire (PVQ-21): A validated short instrument for assessing Schwartz values, used to measure personal values in the study.
- Open-Weight Large Language Models (LLMs): Machine learning models that are trained to follow prescribed value profiles and are evaluated for their sensitivity to situational modifiers in decision-making scenarios.
- Person-Situation Debate: A concept in social psychology that explores whether stable traits or situational contexts better predict behavior, highlighting the importance of both personal values and immediate contextual pressures.

- Modifier-Type Pressure Analysis: An analysis that categorizes situational modifiers into pressure types such as stakes, affective, personal-cost, and informational, to understand their impact on decision-making.
- Profile-Strength Moderation: The concept that the strength of a value profile, determined by the number of active values, can influence the likelihood of decision flips, with stronger profiles being less susceptible to modifier-induced changes.

Important Notations

Definition	Notation
A binary state representing low or high value in Schwartz's value dimensions.	$\{0, 1\}$
A 10-dimensional vector representing Schwartz values for a given action.	$V_{▯ i}$ ▯ $▯^{10}$
A vector representing Schwartz values in a binary configuration.	$V_s w$ ▯ $\{0, 1\}^{10}$
A perturbation vector introduced by a situational modifier to influence decision-making.	δ_m ▯ $▯^{10}$
A utility function representing the value of an action given a Schwartz value profile.	$U(A_i \mid V_s w) = V_s w$▯$V_{▯ i}$
A modifier-conditioned utility function incorporating situational influence.	$U_m(A_i) = V_s w$▯$V_{m ▯ i}$
A decision-flip indicator function showing change in action selection due to situational modifiers.	$F_s(V_s w, S, M) = I[f_s(V_s w, S,$ ▯$) \neq f_s(V_s w, S, M)]$

Literature Review

Literature Review Analysis

Search Keywords for Cited Papers

- Schwartz value theory and PVQ-21
- situational modifiers and person-situation dynamics
- LLM moral decision-making and value steering
- context sensitivity and framing effects
- benchmarking across humans, LLMs, and utility baselines

Most Important Cited Papers

- Gemma Team et. al., Gemma: Open models based on gemini research and technology, 2024: Cited as one of the evaluated LLM families; the Gemma open-weight models form a core experimental baseline for comparing situational flip rates against humans and a utility baseline.
- Bai et. al., Qwen technical report, 2023: Provides the technical foundation for the Qwen models, which are evaluated in the benchmark to assess context-driven decision flips under fixed value profiles.
- Touvron et. al., LLaMA: Open and efficient foundation language models, 2023: Establishes the LLaMA family; a LLaMA model is used as an evaluated baseline to measure scale effects and sensitivity to situational modifiers.
- Schwartz et. al., Refining the theory of basic individual values, 2012: Underpins the construction of fixed value profiles used throughout the evaluation; the study relies on this framework to hold values constant while varying contexts.
- Schwartz et. al., Universals in the content and structure of values: Theoretical advances and empirical tests in 20 countries, 1992: Provides the foundational value theory adopted to align profiles across LLMs and humans in the VISDA experiments and the utility baseline.
- Schwartz, A proposal for measuring value orientations across nations. Questionnaire Package of the European Social Survey, 2003: Cited for PVQ-21, the validated instrument informing how value profiles are measured and operationalized in the study's methodology.

Timeliness of Cited Papers

- The review includes recent post-2024 works central to context sensitivity and value steering, e.g., Guo et al. (2026), Harry et al. (2026), Sauter and Schirmer (2026), Backmann et al. (2025), and Nabizadeh et al. (2026).
- It also cites 2022-2023 studies foundational to priming and value-profiling in LLMs, such as Abdulhai et al. (2023), Bai et al. (2023), Durmus et al. (2023), Kang et al. (2023), and Santurkar et al. (2023).
- Recent analyses of model sensitivity to demographic and geographic priming (Santurkar et al., 2023; Durmus et al., 2023) are cited to motivate context-aware evaluation, aligning VISDA with current concerns about situational dependence in LLM outputs.
- The review integrates recent value-steering and profiling work in LLMs (Guo et al., 2026; Jiao et al., 2025; Kang et al., 2023; Sorensen et al., 2024), ensuring the study's objectives are framed against the latest techniques that manipulate or assess value adherence.
- Contemporary discussions on person-situation dynamics in LLMs (Harry et al., 2026; Soni et al., 2025; Peterson, 2025) are incorporated, indicating awareness of evolving theoretical perspectives in alignment and behavioral variability.

Relevance of Cited Papers

- Direct linkage to value theories: The manuscript grounds its design in Schwartz's value theory and PVQ-21, explaining why Schwartz (Schwartz, 1992; Schwartz et al., 2012; Schwartz, 2003) is chosen over alternatives (e.g., Haidt, 2012) because it maps directly to action-level trade-offs. This establishes a clear theoretical basis for fixed value profiles and justifies the methodological choice of a reduced 10-value representation.
- Clear gap articulation on situational sensitivity: The paper critiques prior LLM value-following work (Sorensen et al., 2024; Abdulhai et al., 2023; Scherrer et al., 2023; Kang et al., 2023; Guo et al., 2026; Jiao et al., 2025) for focusing on whether different value profiles yield different decisions, and positions its contribution as testing whether fixed value profiles yield different actions when situational context changes. This directly supports the objective to introduce VISDA for controlled within-subject tests.
- Integration of social-psychology person-situation evidence: Classic and contemporary findings (Mischel, 1968; Ross and Nisbett, 1991; Harry et al., 2026; Soni et al., 2025; Peterson, 2025) are used to motivate why situational pressures can override stable values, aligning the hypothesis with established evidence that a large share of variance is within-person. This provides strong conceptual justification for measuring decision flips under situational modifiers.
- Use of recent LLM-context literature to justify modifier axes: Empirical demonstrations that contextual priming and framing alter LLM outputs (Santurkar et al., 2023; Durmus et al., 2023; Sauter and Schirmer, 2026; Backmann et al., 2025; Nabizadeh et al., 2026) are explicitly cited to motivate VISDA's modifier categories (e.g., stakes, affective, personal-cost), directly connecting prior observations to the dataset's design and the flip metric.
- Positioning relative to existing moral benchmarks: The manuscript situates VISDA alongside ETHICS (Hendrycks et al., 2021), MoralStories (Emelin et al., 2021), Samway et al. (2025), Mahajan (2025), and Schacht and Lanquillon (2025), noting that these do not systematically vary situational modifiers under fixed value profiles. This comparison clarifies the research gap and contextualizes methodological novelty.
- Methodological relevance through model and human baselines: Inclusion of open-weight LLM families (Gemma Team, 2024; Bai et al., 2023; Touvron et al., 2023) and a human pilot directly supports the study's objective of cross-system comparison under identical value profiles and situational modifiers, reinforcing the empirical scope.

Critical Evaluation

Key Evaluation Points

- The claim that LLMs are generally state-blind is overbroad given cited evidence that situational context and priming shift model outputs. The literature review should reconcile this with findings on context-driven variability. Suggestion: Qualify the state-blindness claim and explicitly integrate evidence from contextual priming and framing studies to position VISDA as testing when and how situational cues override trait prompts.
- The review adopts Schwartz values and notes Haidt's moral foundations only briefly, without discussing implications for generalizability across value frameworks. Suggestion: Add a short comparative paragraph explaining how results might differ under moral foundations and why Schwartz is preferable for action-level operationalization in VISDA, with potential future extensions.
- While the review cites benchmark work (ETHICS, MoralStories, Samway, Mahajan, Schacht and Lanquillon), it stops short of a direct methodological comparison that clarifies exactly how VISDA's within-subject, fixed-profile design advances prior evaluations. Suggestion: Insert a concise table-style narrative comparing VISDA's fixed-profile, modifier-controlled setup against each prior benchmark's scope and limitations to sharpen the novelty claim.

- Operationalization of PVQ-21 for LLM prompting is referenced but not elaborated in the review, leaving unclear how human-measured value constructs map onto model instructions. Suggestion: Briefly describe the mapping from PVQ-21-derived value vectors to prompt templates used to hold value profiles fixed for LLMs and humans, noting any limitations.
- The review emphasizes open-weight families (Gemma, Qwen, LLaMA) but does not discuss whether findings generalize to closed-source models, despite citing evidence about closed-source model agreement. Suggestion: Add one to two sentences discussing how closed-source model agreement observed in prior work relates to expectations for situational sensitivity and VISDA generalizability.

Recommended Papers to Read and Cite

Recent Publications to Improve Timeliness

- Oscar Smallenbroek et al., Constructing Schwartz values framework using the Rokeach values survey: Human value measurement in the longitudinal internet survey for social sciences, 2025. *Link*
- Yinpei Su et al., Understanding How Value Neurons Shape the Generation of Specified Values in LLMs, 2025. *Link*
- Andrey Elster et al., Personal Values and Cognitive Biases, 2024. *Link*
- Nadav Borenstein et al., Investigating Human Values in Online Communities, 2024. *Link*
- Vera Sorin et al., Socio-Demographic Modifiers Shape Large Language Models' Ethical Decisions, 2025. *Link*
- Sanqing Qu et al., LEAD: Learning Decomposition for Source-free Universal Domain Adaptation, 2024. *Link*
- Lauren E. Mullenbach et al., Values outweigh political ideology when forming beliefs about gentrification: A study of the US general public, 2025. *Link*
- Zhiqi Yu et al., Dynamic Target Distribution Estimation for Source-Free Open-Set Domain Adaptation, 2025. *Link*
- A. Sandibekova et al., Pedagogical management to improve environmental education in the context of sustainable development, 2024. *Link*
- Mingxuan Xia et al., A Separation and Alignment Framework for Black-Box Domain Adaptation, 2024. *Link*

Relevant Papers to Address Research Gaps

- Andrea Polonioli, Moving LLM evaluation forward: lessons from human judgment research, 2025. *Link*
- Gaurav Suri et al., Do Large Language Models Show Decision Heuristics Similar to Humans? A Case Study Using GPT-35, 2023. *Link*
- Mohna Chakraborty et al., Structured Moral Reasoning in Language Models: A Value-Grounded Evaluation Framework, 2025. *Link*
- Naama Rozen et al., Do LLMs have Consistent Values?, 2024. *Link*
- Ya Wu et al., The Staircase of Ethics: Probing LLM Value Priorities through Multi-Step Induction to Complex Moral Dilemmas, 2025. *Link*
- Vera Sorin et al., Socio-Demographic Modifiers Shape Large Language Models' Ethical Decisions, 2025. *Link*
- Yu Ying Chiu et al., DailyDilemmas: Revealing Value Preferences of LLMs with Quandaries of Daily Life, 2024. *Link*
- D. Hadar-Shoval et al., Embedded values-like shape ethical reasoning of large language models on primary care ethical dilemmas, 2024. *Link*
- Yinpei Su et al., Understanding How Value Neurons Shape the Generation of Specified Values in LLMs, 2025. *Link*
- D. Hadar-Shoval et al., Assessing the Alignment of Large Language Models With Human Values for Mental Health Integration: Cross-Sectional Study Using Schwartz's Theory of Basic Values, 2024. *Link*

Methodology & Solution

Methodology Analysis

Strengths

- Within-subject pairing for models is correctly implemented: for each (VSW, S, M) tuple, the modified trial is paired with the same model's baseline decision on the identical scenario and value profile to compute Fs, yielding 7,600 paired comparisons per model and aligning with the study goal of isolating situational effects under fixed values.

- Dataset is systematically structured: 5 themes with latent value tensions, 10 scenarios (two per theme) each with A0/A1, and 8 modifier axes (self_preservation, resource_scarcity, social_visibility, in_out_group, time_pressure, diffused_responsibility, competence_uncertainty, authority_signal). Modifier texts are inserted without repeating action labels or adding third options, and a pilot excluded one-sided or overly transparent items. Expected_value_pressure annotations were consensus-resolved and used only in the mathematical baseline.
- Value representation is explicit and curated: 95 binary Schwartz profiles are derived by pruning antagonism-violating combinations, then sampling by profile strength and quadrants to maximize Hamming diversity and balance marginals; natural-language profile descriptions (10 sentences per profile) are used for prompting.
- Transparent, deterministic LLM evaluation: three open-weight instruction-tuned LLMs (LLaMA 3.1 8B, Gemma 4 31B, Qwen 2.5 32B) are run locally on A100 GPU nodes with greedy decoding (temperature=0), and the output space is constrained to a single-token choice (A0/A1), improving reproducibility and label consistency.
- Mathematical baseline is specified with equations and components: action vectors VAi are manually specified over Schwartz dimensions; modifiers add δm with λ=0.5 to form V^m_Ai; decisions flip when argmax utilities change, enabling a mechanistic comparator to non-additive LLM behavior.
- Noise-control effort is present: a paraphrase noise floor uses four meaning-preserving paraphrases for a stratified sample of 95 baseline cells per model, holding values, actions, and decoding fixed to distinguish modifier-induced flips from prompt instability.
- Appropriate inferential testing at the axis level: McNemar's exact test on paired baseline-versus-modifier outcomes with Benjamini-Hochberg FDR across axes supports claims about directional flips per axis.
- Human data collection includes standard psychometrics and controls: PVQ-21 with participant-mean centering, six forced-choice value trade-offs, and SDS-10 for social-desirability screening strengthen construct validity of value measurement before scenario decisions.

Limitations

- Unacknowledged: Ambiguity in scenario counts and totals: the abstract and Section 3.2 describe a dataset of 80 binary moral scenarios with modifiers, yet Section 3.2 specifies 10 base scenarios with 8 modifiers each; Section 3.5 computes modified decisions as 3 × 80 × 95 but baseline decisions as 3 × 10 × 95, creating inconsistency in what constitutes a "scenario" versus a "scenario-modifier pair."
- Unacknowledged: Human study does not mirror the within-subject design: each participant saw a scenario either in baseline or with the modifier that caused the strongest LLM flip for that scenario, but never both; only ~500 decisions were collected, reducing power for axis-level comparisons and coupling the human evaluation to model outcomes.
- Unacknowledged: Utility baseline transparency and sensitivity: VAi and δm are manually specified and λ is fixed at 0.5 without releasing vectors or ablation on λ, leaving baseline conclusions dependent on unvalidated parameter choices.
- Unacknowledged: Reproducibility gaps: prompts, full list of 95 profiles, and Appendix materials are referenced but not provided (Appendix ?? placeholders), preventing independent replication of core components.
- Unacknowledged: Paraphrase robustness is assessed only for baseline scenarios and omits modifier conditions; paraphrase generation criteria and sources are unspecified, leaving uncertainty about modifier-induced flips' sensitivity to wording.
- Unacknowledged: Mapping from PVQ-21 scores to the binary VSW profiles for human-model comparisons is not reported beyond PVQ mean-centering, obscuring how continuous human values were binarized or matched to model profiles.
- Unacknowledged: Decoding-variability not examined: LLMs are evaluated only with greedy decoding at temperature=0; no analyses assess whether modifier sensitivity varies with sampling (e.g., temperature or top-p), limiting reliability assessment across plausible deployment settings.
- Acknowledged: Only open-weight models are evaluated; closed-weight frontier models (GPT-4, Claude, Gemini) could show different patterns, limiting generality.
- Acknowledged: Convenience human sample from a single educational and cultural context limits generalizability; authors call for broader replication.
- Acknowledged: Use of the 10-value Schwartz taxonomy instead of the refined 19-value theory may obscure finer distinctions and value-specific modifier effects.
- Acknowledged: Modifiers authored by the research team across eight axes may introduce selection bias; a crowdsourced or LLM-augmented, adversarially validated process is suggested.
- Acknowledged: Binary encoding of values coarsens continuous constructs and pruning antagonism-violating combinations constrains the profile space.

Proposed Solution Analysis

Strengths

- Clear problem framing and metric: VISDA formalizes scenario-modifier design and defines Fs to quantify situationally induced decision changes under fixed values; system-level flip rates are computed over all valid (VSW, S, M) tuples.
- Triangulated evaluation across systems: the study compares a utility baseline, three open-weight LLMs, and a 50-participant human pilot using the same value framework; the utility baseline flips on 19.93% of trials while LLMs show 2-9% and humans nontrivial flips, establishing comparative context sensitivity.
- Granular axis-level and scenario-level analyses: per-axis flip rates with FDR-controlled McNemar tests show authority_signal and self_preservation as top axes for all LLMs; per-scenario heatmaps identify SC004_2 + self_preservation as the strongest cell (42.1% Gemma, 31.6% Qwen), revealing clustered sensitivity.
- Cross-model consistency strengthens claims: identical top-two axes across Gemma, Qwen, and LLaMA with moderate Spearman correlations (e.g., ρ=0.66 between ≥30B models) reduce the likelihood of a single-lab artifact and distinguish behavior from the additive baseline.
- Non-additive contextual reasoning is evidenced: LLMs execute "impossible" flips under strong profiles where the additive baseline flips zero by construction, indicating integration of context beyond linear value arithmetic; formal conditions on utility margins are stated.
- Modifier-type pressure analysis reveals directional miscalibration: humans are most sensitive to stakes (16.3%) and affective (15.3%), whereas Gemma and Qwen emphasize personal-cost (notably self_preservation) and under-react to stakes; LLaMA 8B shows ≤2.6% across types, providing interpretable, category-level gaps.
- Operational scale and efficiency: the pipeline yields 22,800 modified and 2,850 baseline LLM decisions across models with single-token outputs, demonstrating practical feasibility on standard research compute.

Limitations

- Unacknowledged: Effectiveness claims rely primarily on flip rates without reporting uncertainty for cross-system comparisons (e.g., CIs for human-LLM gaps or between-model differences) beyond per-axis McNemar tests; effect sizes at the system level are not quantified.
- Unacknowledged: Human evaluation is partially model-coupled by selecting, per scenario, the modifier producing the strongest LLM flip, risking overestimation of divergence and limiting an unbiased assessment across the full modifier set.
- Unacknowledged: Limited coverage and inconsistent scenario accounting may constrain generalizability and scalability claims: only five themes and ten base scenarios are included, while the text alternates between 10 scenarios and 80 scenario-modifier pairs.
- Unacknowledged: Baseline comparisons depend on manually set VAi, δm, and λ=0.5 without ablations or release of these parameters, reducing robustness of claimed advances over additive models.
- Acknowledged: Generality is limited to open-weight models; closed-weight frontier models might alter axis-sensitivity conclusions.
- Acknowledged: Coarse value granularity (10-value taxonomy, binary encoding) and team-authored modifiers may bias coverage and obscure fine-grained effects; authors suggest refined taxonomies and broader modifier generation in future work.

Critical Evaluation

Key Evaluation Points

- Scenario accounting is inconsistent between "scenarios" and "scenario-modifier pairs," and baseline versus modified totals are computed with different denominators. Suggestion: Standardize terminology and enumerate all 10 base scenarios and 80 scenario-modifier pairs in a public manifest; align all totals to the same unit of analysis.
- Human evaluation is not within-subject and is tied to model-selected modifiers, limiting unbiased human-LLM comparison. Suggestion: Re-run with within-subject or balanced between-subjects designs across randomly assigned modifiers (e.g., Latin square), avoiding selection based on LLM flips.
- Utility baseline parameters (VAi, δm, λ=0.5) are manually specified without release or sensitivity analysis. Suggestion: Publish the vectors and run ablations varying λ and alternative encodings (e.g., crowd-informed), reporting stability with confidence intervals.

- Core materials for replication (prompts, full profile list, paraphrase details) are missing due to Appendix placeholders. Suggestion: Release prompt templates, the 95 value profiles with natural-language descriptions, paraphrase generation criteria, and random seeds in a repository.
- Paraphrase robustness is only measured for baseline items and lacks generation details, leaving modifier sensitivity untested. Suggestion: Extend paraphrase evaluations to modified scenarios across axes and report flip-rate changes with confidence intervals.
- Mapping from continuous PVQ-21 responses to binary VSW profiles is unspecified. Suggestion: Specify and validate the binarization/matching procedure (e.g., thresholding or nearest-profile mapping) and report agreement with continuous measures.
- Only greedy decoding is tested, so flip sensitivity under stochastic decoding remains unknown. Suggestion: Evaluate across temperatures/top-p settings and report stability of axis rankings and flip rates.
- Cross-system differences lack uncertainty quantification beyond per-axis tests. Suggestion: Report system-level effect sizes and confidence intervals for human-LLM and between-model gaps.
- Limited thematic and scenario coverage constrains generalizability. Suggestion: Expand themes, scenarios, and modifiers beyond the current five themes and ten base scenarios, and document selection criteria to improve breadth.

Recommended Papers to Read and Cite

Relevant Papers to Strengthen Methodology

- Chris Shaner et al., Assessing Moral Decision Making in Large Language Models, 2025. *Link*
- Vanessa Cheung et al., Large language models show amplified cognitive biases in moral decision-making, 2025. *Link*
- Vera Sorin et al., Socio-Demographic Modifiers Shape Large Language Models' Ethical Decisions, 2025. *Link*
- Mohna Chakraborty et al., Structured Moral Reasoning in Language Models: A Value-Grounded Evaluation Framework, 2025. *Link*
- Karina Vida et al., Decoding Multilingual Moral Preferences: Unveiling LLM's Biases Through the Moral Machine Experiment, 2024. *Link*
- M. Ahmad et al., Large-scale moral machine experiment on large language models, 2024. *Link*
- Andrea Polonioli, Moving LLM evaluation forward: lessons from human judgment research, 2025. *Link*
- Pratik S. Sachdeva et al., Normative Evaluation of Large Language Models with Everyday Moral Dilemmas, 2025. *Link*
- José Luiz Nunes et al., Are Large Language Models Moral Hypocrites? A Study Based on Moral Foundations, 2024. *Link*
- Junfeng Jiao et al., LLM ethics benchmark: a three-dimensional assessment system for evaluating moral reasoning in large language models, 2025. *Link*

Results & Findings

Results & Findings Analysis

Strengths

- Explicit experimental control aligns with the research question: values are held constant across 95 Schwartz profiles while situational modifiers vary across 80 scenario–modifier pairs, yielding 7,600 modified trials per LLM and enabling a direct test of whether situational factors drive decision flips beyond values. Flip rates show the utility baseline at 19.93% vs LLMs at 2.03%-8.92% and a human pooled shift of 12.36% (N=50, 500 decisions), directly addressing the hypothesis.
- Paraphrase noise is quantified and clearly separated from modifier effects: four meaning-preserving paraphrases are applied to a stratified sample of 95 baseline (VSW, scenario) cells per model while holding value profile, candidate actions, response format, and decoding settings fixed; paraphrase-induced flips are 2.95% (Gemma), 2.64% (Qwen), and 0.00% (LLaMA 8B), all significantly below modifier-induced flip rates by two-proportion z-tests ($z > 4.5$, $p < 1e-5$). Paraphrase flips localize to a single scenario per model, whereas modifier flips occur across all ten scenarios.
- Per-axis analysis is statistically grounded: McNemar's exact test with Benjamini-Hochberg FDR correction across the eight axes is used to assess directional flips for each (model, axis) cell, with authority_signal and self_preservation

jointly accounting for roughly 30-40% of all LLM reversals. Humans peak on in_out_group, authority_signal, and resource_scarcity, highlighting axis-specific divergences.

- Cross-model replication of axis ordering: the top two axes by flip rate are authority_signal and self_preservation across Gemma 4 31B, Qwen 2.5 32B, and LLaMA 3.1 8B; Spearman ρ on the eight-axis ordering is 0.66 between Gemma and Qwen ($p = 0.076$), with weaker correlations involving LLaMA 8B (Qwen $\rho = 0.49$, Gemma $\rho = 0.18$). The additive baseline is weakly or negatively correlated with LLMs (ρ ▯ [-0.13, 0.35]).
- Fine-grained per-scenario evidence clarifies concentration of effects: heatmaps show concentration on self_preservation and authority_signal; the strongest cell is SC004_2 + self_preservation with flip rates of 42.1% (Gemma) and 31.6% (Qwen). This pinpoints scenario–axis combinations driving reversals rather than diffuse, uniform effects.
- Profile-strength moderation reveals non-additive reasoning: the utility baseline flips 0% for strength-8/9 profiles by construction, whereas Gemma flips 20.3% (strength-8) and 15.6% (strength-9), Qwen 10.3% and 5.0%, and LLaMA 0.6% and 0.0%; the formal proposition shows such flips are impossible under the additive value-vector rule, indicating context integration beyond value addition.
- Modifier-type analysis provides concrete human-LLM miscalibration targets: humans show largest mean decision shifts for stakes (16.3%) and affective (15.3%), while Gemma and Qwen peak on personal-cost (9.7% and 8.9%) and under-react to stakes; LLaMA 8B is nearly flat across types (≤2.6%). Table 2 and Figure 4 quantify these differences.
- Clarity in presentation: Table 1 reports exact flip counts and rates; Figure 2 provides per-axis comparisons across systems; Figure 3 visualizes scenario-axis heatmaps; Figure 4 and Table 2 summarize modifier-type effects, enabling multi-level inspection of results.
- Interpretation ties to prior literature: the dominant LLM axes align with classic situational factors of obedience and self-interest; conclusions emphasize that LLMs are not purely additive and that axis-level calibration is needed given humans' higher sensitivity to material stakes and in-group/authority framing.

Limitations

- Acknowledged: Model scope restricted to open-weight LLMs; closed-weight frontier models (GPT-4, Claude, Gemini) might show different scale and modifier patterns that could alter the reported human-LLM divergence.
- Acknowledged: Sample diversity is limited to a convenience sample from a single educational/cultural context; broader replication is needed for generality.
- Acknowledged: Value taxonomy uses the 10-value Schwarz framework rather than the refined 19-value theory, potentially collapsing distinctions (e.g., humility, face, types of conformity) and masking value-specific modifier effects.
- Acknowledged: Modifiers were authored by the research team across 8 axes, introducing selection bias; future work suggests crowdsourced or LLM-augmented generation with adversarial validation.
- Acknowledged: Binary value representation and pruning of antagonism-violating combinations coarsen signal and constrain the profile space relative to continuous real-world values.
- Acknowledged: Human evaluation is between-subjects; within-trial flips are undefined, and the reported human metric is pooled $|\Delta P(A1)|$ across 10 scenario-axis cells, limiting direct comparability to model within-trial flips.
- Unacknowledged: Incomplete statistical reporting: McNemar's exact test results and the full profile-strength breakdown are deferred to Appendix ??, which is not provided; Table 2 and Figure 4 lack statistical significance tests for between-system comparisons.
- Unacknowledged: Limited breadth of human coverage: only 10 scenario-axis cells are covered via the two-form rotation, while LLMs are evaluated on the full 80 scenario–modifier pairs; coverage balance and its impact on axis-level comparisons are not quantified.
- Unacknowledged: Decoding hyperparameters are unspecified beyond stating that response format and decoding settings were fixed, hindering reproducibility of the flip-rate noise floor and modifier effects. Exact values for temperature, top-p, and max tokens are not reported.
- Unacknowledged: Scenario confounds are plausible: paraphrase flips localize to a single scenario per model, and strong per-scenario concentration (e.g., SC004_2 + self_preservation) suggests high-leverage cells may disproportionately influence axis-level rates without per-scenario normalization.
- Unacknowledged: The manuscript claims a clear scale effect in flip rates but provides no formal trend analysis or regression quantifying the relationship between model size and flip rate.

Comparative Analysis against Cited Papers

Improvements

- Decision-flip metric isolates situational sensitivity beyond values and improves realism relative to an additive value-arithmetic baseline: utility baseline flips at 19.93% vs LLMs at 2.03%-8.92%, with the human pooled shift at 12.36% between these, indicating non-additive contextual integration by learned models.
- Cross-model convergence on axis sensitivity absent in the additive baseline: authority_signal and self_preservation are top-ranked across all three LLMs; Spearman $\rho = 0.66$ between Gemma and Qwen ($p = 0.076$), whereas the baseline's axis ordering correlates weakly or negatively with LLMs ($\rho \in [-0.13, 0.35]$).
- Paraphrase control strengthens causal attribution to modifiers: paraphrase noise is significantly lower than modifier-induced flips across models ($z > 4.5$, $p < 1e-5$), and paraphrase flips localize to single scenarios per model, ruling out surface-form instability as the main driver.
- Modifier-type analysis reveals actionable miscalibration targets relative to humans: humans prioritize stakes (16.3%) and affective (15.3%) while Gemma and Qwen prioritize personal-cost (9.7%, 8.9%) and under-react to stakes; LLaMA 8B shows uniformly low sensitivity (≤2.6%).

Contradictions

- Contradicts the assumption that fixed value profiles reliably predict behavior: with values held constant, situational modifiers cause meaningful flips in LLMs (2.03%-8.92%) and humans (12.36% pooled), demonstrating values alone are insufficient.
- A simple additive baseline aligns better with human per-axis ordering ($\rho = +0.60$) than any evaluated LLM ($\rho = +0.16$ Gemma, -0.05 Qwen, $+0.29$ LLaMA 8B; all $p > 0.1$), contrary to expectations that learned models should match human sensitivity more closely than a hand-specified utility rule.

Critical Evaluation

Key Evaluation Points

- Incomplete statistical reporting for McNemar tests, profile-strength breakdowns, and no significance testing for modifier-type comparisons. Suggestion: Provide full McNemar statistics with Benjamini-Hochberg-adjusted p-values, per-axis effect sizes with confidence intervals, the complete profile-strength table referenced as Appendix, and statistical tests comparing modifier-type shifts across systems.
- Human coverage is narrow (10 scenario-axis cells) relative to full LLM coverage, limiting comparability of axis-level conclusions. Suggestion: Expand the human study to cover all 80 scenario–modifier pairs or implement a mixed within-between design enabling within-participant baselines, and report coverage-weighted per-axis estimates.
- Decoding hyperparameters are unspecified, limiting reproducibility of noise floors and flip rates. Suggestion: Report exact prompts and decoding hyperparameters (temperature, top-p, max tokens) and include a sensitivity analysis showing flip-rate stability across decoding settings.
- Scenario-specific concentration may confound axis-level rates (e.g., SC004_2 + self_preservation dominates). Suggestion: Use per-scenario normalization or hierarchical models to separate scenario and axis effects, and report robustness when excluding high-leverage scenarios.
- Scale effect is claimed without formal trend quantification. Suggestion: Fit and report a regression of flip rate on model size and, if possible, include additional intermediate model scales to quantify the relationship and its uncertainty.
- LLMs align worse with human per-axis ordering than the additive baseline. Suggestion: Explore calibration of axis weights via human-labeled importance scores or targeted fine-tuning to increase sensitivity to stakes and affective modifiers, then reassess Spearman rank alignment.

Recommended Papers to Read and Cite

Relevant Papers to Address Gaps in Findings

- Vera Sorin et al., Socio-Demographic Modifiers Shape Large Language Models' Ethical Decisions, 2025. *Link*
- Mohna Chakraborty et al., Structured Moral Reasoning in Language Models: A Value-Grounded Evaluation Framework, 2025. *Link*
- Ya Wu et al., The Staircase of Ethics: Probing LLM Value Priorities through Multi-Step Induction to Complex Moral Dilemmas, 2025. *Link*
- J. Carpendale et al., Situational variation in moral judgment: In a stage or on a stage?, 1992. *Link*

- A. Cadiente et al., Large Language Models Take on the AAMC Situational Judgment Test: Evaluating Dilemma-Based Scenarios, 2025. *Link*
- Heena Rathore et al., Assessing Gender and Age Influences on Moral Decision Making in Autonomous Vehicles Using Large Language Models, 2025. *Link*
- Sylvie Delacroix, Moral Perception and Uncertainty Expression in LLM-Augmented Judicial Practice, 2025. *Link*
- Patricia L. Lockwood et al., Moral Learning and Decision-Making Across the Lifespan, 2024. *Link*
- Elyas Barabadi et al., Comparing AI and human moral reasoning: context-sensitive patterns beyond utilitarian bias, 2026. *Link*
- M. Ahmad et al., Large-scale moral machine experiment on large language models, 2024. *Link*

Concise Report

Literature Review

Strengths

- Clear theoretical grounding in Schwartz's value theory and PVQ-21, with an explicit rationale for selecting the 10-value representation to map directly to action-level trade-offs, contrasted against moral foundations theory.
- The study explicitly targets a documented gap: prior LLM work primarily varies value profiles, whereas this paper holds value profiles fixed and varies situational context to test for decision flips, motivating the VISDA benchmark.
- Integration of person-situation research from social psychology to justify context-driven behavior variance, including classic sources and recent within-person variance evidence, directly aligning with the decision-flip metric.
- Use of recent LLM-context literature (priming, framing, survival risk, opponent behavior) to justify VISDA's modifier axes and their expected pressures, linking contextual effects to moral decision changes under fixed values.
- Positioning against moral reasoning benchmarks (ETHICS, MoralStories, Samway, Mahajan, Schacht and Lanquillon) to emphasize that existing resources lack systematic variation of situational modifiers under fixed value profiles.
- Methodological relevance is reinforced by evaluating three open-weight LLM families (LLaMA 3.1 8B, Gemma 4 31B, Qwen 2.5 32B) and a human pilot within the same value framework and scenario-modifier design.

Limitations & Suggestions

- Overbroad characterization that LLMs are state-blind despite cited evidence of contextual priming effects. Suggestion: Qualify the claim and explicitly integrate priming/framing findings to position VISDA as quantifying when situational cues override trait-level value prompts.
- Limited discussion of generalizability across value frameworks beyond briefly noting Haidt's moral foundations. Suggestion: Add a focused comparison outlining how results might differ under moral foundations and why Schwartz's values better support action-level operationalization here.
- Novelty over prior benchmarks is asserted without a direct methodological contrast. Suggestion: Insert a concise comparative narrative contrasting VISDA's fixed-profile, modifier-controlled within-subject design with ETHICS, MoralStories, and recent moral reasoning analyses.
- PVQ-21 operationalization for LLM prompting is referenced but mapping details are omitted. Suggestion: Briefly detail how PVQ-21 constructs inform the natural-language profile descriptions used in prompts and any limits of this mapping.
- Generalization to closed-weight models is not discussed though prior work notes closed-source agreement. Suggestion: Add one to two sentences on how reported closed-source alignment findings inform expectations for situational sensitivity and VISDA's applicability.

Methodology & Solution

Strengths

- Within-subject pairing is implemented per system: each modified trial (VSW, S, M) is paired with the same model's baseline decision on the same (VSW, S) to compute Fs, yielding 7,600 paired comparisons per model.
- Dataset structure is explicit and controlled: 5 themes, 10 scenarios (two per theme) each with actor and A0/A1, and 8 modifier axes (self_preservation, resource_scarcity, social_visibility, in_out_group, time_pressure,

diffused_responsibility, competence_uncertainty, authority_signal); wording normalization, lexical leakage checks, and pilot filtering are described.

- Value profiles: 95 curated binary Schwartz configurations are constructed by pruning antagonism-violating combinations, then sampling to maximize Hamming diversity and balanced marginals; each profile has a 10-sentence description for prompting.
- Deterministic LLM evaluation: LLaMA 3.1 8B, Gemma 4 31B, and Qwen 2.5 32B are run locally with greedy decoding (temperature = 0), single-token outputs (A0/A1), and 22,800 modified plus 2,850 baseline decisions across models.
- Utility baseline is formally specified: U(Ai|Vsw)=Vsw^T VAi with modifier-adjusted V^m_Ai = VAi + λδm (λ=0.5), and flips recorded when argmax utilities change; δm derives from expected_value_pressure annotations.
- Paraphrase noise control: four meaning-preserving paraphrases are tested for a stratified sample of 95 baseline (VSW, S) cells per model, holding profiles, actions, and decoding fixed, to separate prompt instability from modifier effects.
- Axis-level inference uses McNemar's exact test on paired outcomes per (model, axis), with Benjamini-Hochberg FDR correction across eight axes.
- Human study includes standardized psychometrics: PVQ-21 with participant-mean centering, six forced-choice value trade-offs, and SDS-10 social-desirability screen before scenario decisions.

Limitations & Suggestions

- Scenario accounting is inconsistent: the paper alternates between 80 binary moral scenarios and 10 base scenarios with 8 modifiers; LLM modified totals use 80 while baseline totals use 10, obscuring units of analysis. Suggestion: Standardize terminology and publish a manifest mapping the 10 base scenarios to 80 scenario-modifier pairs; align denominators across sections.
- Human evaluation is model-coupled and not within-subject: each participant saw baseline or the modifier producing the strongest LLM flip for that scenario only once, reducing unbiased axis-level comparison and power. Suggestion: Re-run with within-subject or balanced between-subjects designs using randomly assigned modifiers independent of LLM flips.
- Utility baseline depends on manually set VAi, δm, and λ=0.5 without released vectors or sensitivity analyses. Suggestion: Publish VAi and δm, and ablate λ and alternative encodings to assess robustness with confidence intervals.
- Core replication materials are missing (Appendix placeholders for profiles, prompts, and analysis details). Suggestion: Release the full list of 95 profiles with natural-language descriptions, prompt templates, and seeds in a repository.
- Paraphrase robustness is evaluated only for baseline cells, not modified scenarios; paraphrase generation criteria are unspecified. Suggestion: Extend paraphrase tests to modified prompts across axes and document paraphrase generation methods.
- Mapping from continuous PVQ-21 scores to binary VSW profiles for human-model comparisons is unspecified. Suggestion: Report binarization or nearest-profile mapping procedures and validate against continuous measures.
- Only greedy decoding (temperature = 0) is assessed for LLMs; stochastic decoding effects on flip sensitivity are unknown. Suggestion: Evaluate temperature/top-p variations to test stability of flip rates and axis rankings.
- Evaluation excludes closed-weight models, limiting generality. Suggestion: Where feasible, include GPT-4-class models or explicitly discuss expected differences to contextualize generalizability.
- Value representation and modifier authorship may bias coverage: 10-value binary encoding and team-authored modifiers can obscure fine-grained effects. Suggestion: Explore the refined 19-value taxonomy and diversify modifier generation via crowdsourcing or LLM-augmented, adversarial validation.

Results & Findings

Strengths

- System-level flip rates directly quantify situational sensitivity under fixed values: utility baseline 1,515 flips (19.93%), Gemma 4 31B 678 (8.92%), Qwen 2.5 32B 500 (6.58%), LLaMA 3.1 8B 154 (2.03%), and human pooled |ΔP(A1)| = 12.36% across 10 scenario-axis cells.
- Paraphrase noise floors are significantly below modifier-induced flips: 2.95% (Gemma), 2.64% (Qwen), 0.00% (LLaMA 8B) vs their respective modifier flip rates; two-proportion z-tests reject equality ($z > 4.5$, $p < 1e-5$), and paraphrase flips localize to a single scenario per model, unlike modifier flips that span all ten scenarios.
- Per-axis analysis shows authority_signal and self_preservation jointly account for roughly 30-40% of LLM reversals, with McNemar's tests and Benjamini-Hochberg FDR controlling for multiple axes; humans peak on in_out_group,

authority_signal, and resource_scarcity.

- Cross-model convergence: identical top two axes (authority_signal, self_preservation) across all three LLMs; Spearman $\rho = 0.66$ between Gemma 4 31B and Qwen 2.5 32B ($p = 0.076$), with weaker correlations involving LLaMA 3.1 8B; the additive baseline correlates weakly or negatively with every LLM ($\rho \in [-0.13, 0.35]$).
- Per-scenario concentration is identified: SC004_2 + self_preservation is the strongest cell in both Gemma (42.1%) and Qwen (31.6%) heatmaps; flips cluster on self_preservation and authority_signal across scenarios.
- Profile-strength moderation evidences non-additivity: the utility baseline flips 0% for strength-8/9 profiles, whereas Gemma flips 20.3% (strength-8) and 15.6% (strength-9), Qwen 10.3% and 5.0%, and LLaMA 0.6% and 0.0%; a formal proposition shows such flips are impossible under the additive rule, indicating contextual premise integration.
- Modifier-type pressure analysis quantifies human-LLM miscalibration: humans show higher mean shifts for stakes (16.3%) and affective (15.3%) than personal-cost (10.6%) and informational (6.1%); Gemma and Qwen emphasize personal-cost (9.7%, 8.9%) and under-react to stakes; LLaMA 3.1 8B is ≤2.6% across types.
- Presentation is precise and multi-level: Table 1 enumerates flip counts and rates; Figure 2 shows per-axis comparisons; Figure 3 provides scenario-axis heatmaps; Table 2 and Figure 4 summarize modifier-type effects.

Limitations & Suggestions

- Incomplete statistical detail: McNemar's exact test outputs, full profile-strength breakdowns, and between-system tests for modifier-type differences are deferred to missing appendices. Suggestion: Report adjusted p-values, effect sizes with confidence intervals, and include the complete profile-strength table referenced.
- Human coverage is narrow: only 10 scenario-axis cells via two-form rotation vs models evaluated on all 80 scenario-modifier pairs, limiting direct comparability of axis-level patterns. Suggestion: Expand human evaluation to all 80 pairs or implement a mixed within-between design enabling within-participant baselines.
- Decoding hyperparameters beyond temperature=0 are under-specified in results contexts, constraining reproducibility of noise floors and flip rates. Suggestion: Report exact prompts and decoding settings (e.g., top-p, max tokens) and include a sensitivity analysis across sampling regimes.
- Scenario-level leverage may confound axis-level rates (e.g., SC004_2 + self_preservation dominates). Suggestion: Apply per-scenario normalization or hierarchical modeling and test robustness by excluding high-leverage cells.
- A scale effect is claimed but not quantified with a formal trend. Suggestion: Fit a regression of flip rate on model size and include intermediate scales to estimate trend and uncertainty.